

ASHAB ALI KHAN

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Publications

- **Fruit Classification and Calorie Estimation Using Machine Learning (2025)** Accepted & In Press
7th International Conference on Engineering & Advancement in Technology (ICEAT) Co-Author (3-Member Research Team)
 - Engineered the core Convolutional Neural Network (CNN) architecture and executed iterative model training experiments to optimize automated image classification accuracy to 97.96%.
 - Co-authored the peer-reviewed technical manuscript, constructing comprehensive mathematical documentation of the system's underlying calorie estimation matrices.

Technical Skills

- **AI & Machine Learning:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Agentic AI, LangChain, LangGraph, Scikit-learn, TensorFlow, Keras.
- **Programming & Data:** Python, C++, SQL, NumPy, Pandas.
- **NLP & AI Tools:** NLP, Arabic NLP, Embeddings, Sentence Transformers, Semantic Search, Vector Search, ChromaDB, Groq API, Tavily, Tesseract OCR.
- **Tools & Deployment:** Docker, Streamlit, Hugging Face Spaces, Git, GitHub, Google Colab, REST APIs, API Integration.
- **Computer Science Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Software Engineering.

Projects

- **GCC AI & Startup Intelligence Platform (LIVE)** Personal Project
GitHub Repository Jun 2026
 - Orchestrated end-to-end **multi-agent pipeline** using **LangGraph**, managing state passing between **6 sequential agents** from query intake to email delivery.
 - Architected a **6-agent LangGraph pipeline** processing over **30 web sources** across **6 GCC countries**, delivering professional Word reports via email in under **2 minutes**.
 - Constructed a **Query Understanding Agent** capable of classifying and routing **six specific query types**, enabling faster, more targeted research report generation.
 - **Assumed full ownership of scale optimization**; implemented batch processing of 6 sources per LLM call, resolving token limit crashes across the 30+ source data stream.
 - Engineered an **autonomous Review Agent** decision system, filtering **13-18 irrelevant entities** per query, thus enhancing the accuracy and relevance of generated reports.
 - **Expanded project scope from core logic to full-stack deployment**; streamlined the integration of Tavily, Groq LLM, SendGrid, and a Streamlit dashboard via Docker.
- **Arabic RAG Chatbot (LIVE)** Personal Project
GitHub Repository Jan 2026
 - **Conceived and launched** a bilingual **Arabic-English RAG chatbot** for PDF querying using **LangChain, ChromaDB, and Arabic-Triplet-Matryoshka-V2 embeddings**.
 - **Diagnosed execution blockers**; and shifted to **Tesseract OCR** to bypass non-standard font encoding corruption across 81 pages.
 - **Eliminated** cross-lingual retrieval failure by **building an English-to-Arabic translation bridge**, aligning both languages into the same vector space.
 - Engineered **hybrid conversation memory** — last 2 full exchanges plus LLM-generated rolling summary — keeping token cost **constant at scale**.
 - **Deployed on Hugging Face Spaces** via Docker; resolved proxy-level 403 errors by disabling Streamlit XSRF protection in container launch configuration.
 - Implemented **per-message RTL rendering logic** storing language per conversation turn, preventing CSS alignment conflicts across mixed Arabic-English chat history.

Education

- **Galgotias University** Greater Noida, UP
Master of Computer Application (MCA) 2023 - 2025
 - **Core Focus Areas:** Machine Learning, Advanced Data Structures & Algorithms, Database Management Systems (DBMS).
- **Mahatma Jyotiba Phule Rohilkhand University (MJPRU)** Bareilly, UP
Bachelor of Science (BSc) — Physics, Chemistry, & Mathematics 2019 - 2023